

OceanAISpinup Architecture, Design, and Pilot Results

OceanAISpinup — Architecture, Design, and Pilot Results

Audience: team presentations alongside docs/handoff_briefing.html

Branch: handoff/prototype-e2e

Case: v3.GMPAS-NYF_QU240 (E3SM / MPAS-Ocean)

Date: 2026-09-10 (metrics from the published CPU pilot)

This document is the **architecture narrative** for the AI effort: what we learn, why the design looks the way it does, how GraphCast and LandSim relate, how data are prepared, which restart fields are predicted, and what the holdout numbers show (including per-layer temperature).

1. Problem statement

Ocean spinup to a near-equilibrium restart is expensive. LandSim already showed that an AI model can map **early spinup state** → **near-equilibrium restart fields** for ELM. For MPAS-Ocean the same *job* exists, but the state lives on an **unstructured C-grid**, so spatial coupling cannot be ignored.

OceanAISpinup v1 (pilot) learns a **restart operator** on one coarse mesh:

$$[\{t+\} = x\{t\} + f_!(x_ \{t\},, F_ \{ \},, G) \text{ } k = 4660 \text{ }]$$

Symbol	Meaning in the pilot
(x_t)	Early restart (X), years 51–55
(x_{t+})	Late restart (Y), years 601–605 ($\Delta = 550 \text{ yr}$, same calendar month)
(F_)	CORE2 Normal Year Forcing (monthly, remapped to ocean cells)
(G)	QU240 cell dual graph from cells0nCell
(f_)	Residual GraphSAGE (cell-only)

Deliverable: a **legal MPAS-Ocean restart** NetCDF: copy the early file, overwrite only deep temperature and salinity, keep thickness, velocity, mesh, and xtime.

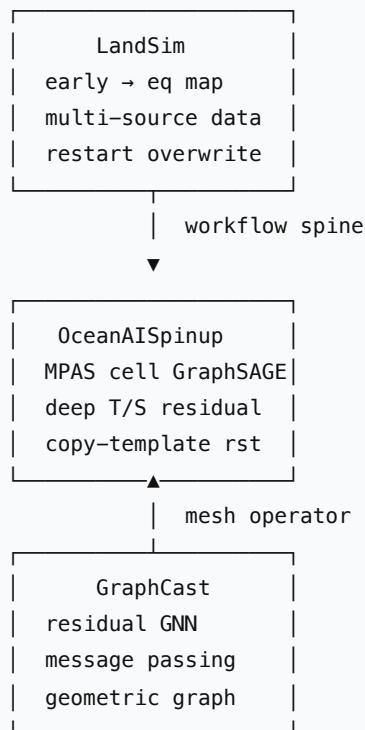
This is **not** a weather forecast, **not** GenCast, and **not** a full-column ocean emulator.

2. Design principles

1. **Restart contract first.** Any prediction must land in a file MPAS can open. Writeback is copy-template overwrite, proven with a truth-Y dry run before trusting the model.
 2. **Native unstructured mesh.** Ocean fields stay on MPAS cells. Do not interpolate T/S to lat-lon for a ViT/UNet and interpolate back.
 3. **Deep ocean only in v1.** Alice/Hyun OHC bands show the 2000 m-bottom layer is the slow spinup signal. Levels with $\text{refBottomDepth} > 2000 \text{ m} \rightarrow \mathbf{k = 46...60}$ (15 levels).
 4. **Residual learning.** Predict (T, S) and add to X. Persistence ((Y = X)) is the baseline every model must beat.
 5. **Area $\times$ thickness weighted loss.** Use areaCell and layer thickness so large deep cells dominate RMSE/OHC consistently with ocean integrals.
 6. **Deterministic GNN before diffusion.** With 48 train pairs on one NYF case, a GenCast-style generative model is the wrong first objective (see `OceanAISpinup_Diffusion_Evaluation.md`).
 7. **Encode atmosphere and namelist, do not over-claim causality.** NYF cycles and `mpaso_in` is fixed for this case — (F) and () do not vary across samples.
 8. **Ship the workflow.** The pilot's value is the end-to-end path (pairs $\rightarrow$ graph $\rightarrow$ train $\rightarrow$ restart $\rightarrow$ X/ML/Y snapshot), not a production GNN.
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3. Relationship to GraphCast and LandSim

OceanAISpinup is a **hybrid of two parents**. It does **not** import either codebase; it reuses their *ideas*.



3.1 What we take from LandSim

LandSim pattern	OceanAISpinup analogue
Early AD-spinup (X) vs final-spinup (Y) pairs	Year ~50 vs ~600 month-aligned restart pairs
Modular dataGEN / IO lists	<code>oceanai.data</code> pair index, extract, DATM encoder, scalers
Per-variable normalization	Train-only z-score on T/S and $\Delta T/\Delta S$
Multi-task weighted loss	Huber on $\Delta T/\Delta S$ with <code>areaXthickness</code> weights
<code>ai_predictions_to_restart</code> copy-template overwrite	<code>oceanai.io.write_restart.write_deep_ts</code>

What we do *not* copy: LandSim’s **per-gridcell IID Transformer**. Land cells are treated as independent samples; deep ocean equilibration is **nonlocal**. A cell-independent model is the wrong inductive bias for MPAS-Ocean.

3.2 What we take from GraphCast

GraphCast pattern	OceanAISpinup analogue
Message passing on a sphere graph	GraphSAGE on MPAS <code>cellsOnCell</code> dual
Predict normalized residual of the next state	$(T = T_X + ())$ (same for S)
Vertical levels as channels	15 deep levels stacked in the feature/target vectors
Area-aware loss (lat weights on ERA5)	<code>areaCell</code> $\times$ <code>layerThickness</code> on the deep mask

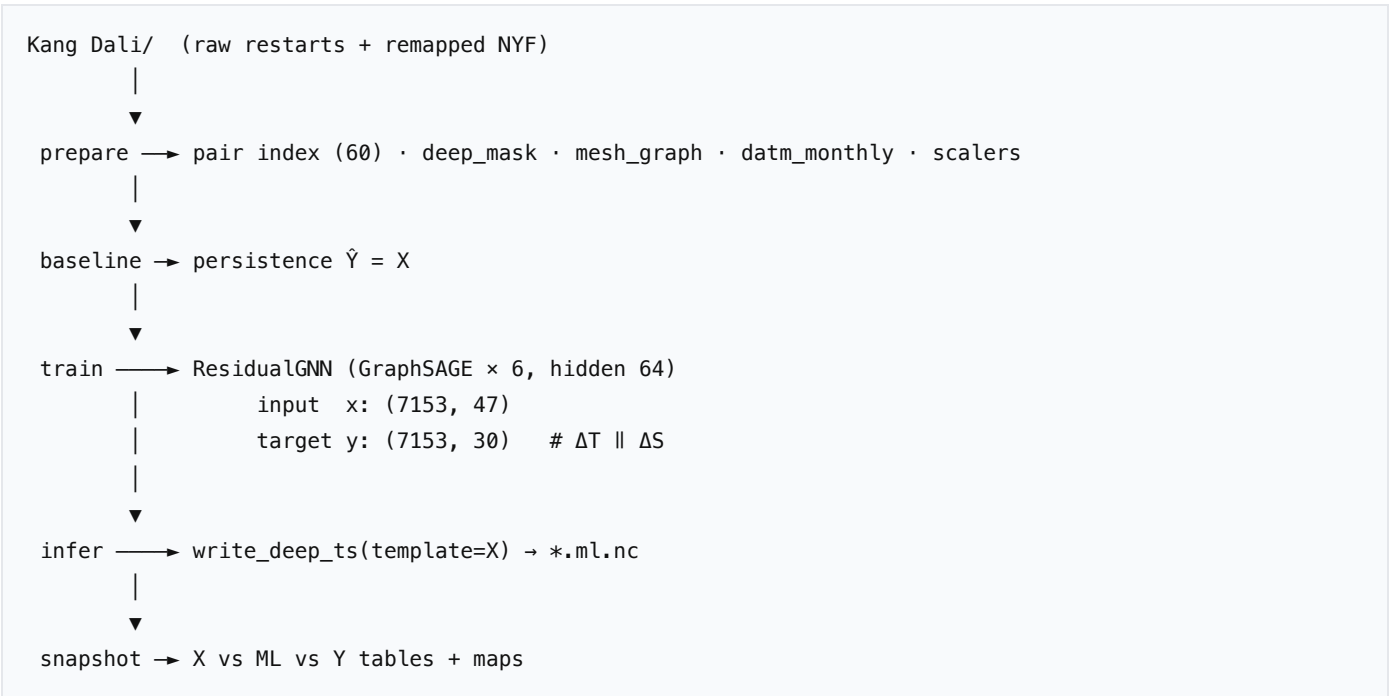
What we do *not* copy:

GraphCast / GenCast piece	Why it does not transfer as-is
Separate icosahedral “dynamics mesh”	MPAS is already the unstructured mesh
Grid↔mesh bipartite encode/decode	Physics already lives on cells; DATM is remapped offline
6-hour weather rollout	We map a 550-year spinup jump
GenCast diffusion / ensembles	Need many independent pairs and proven multi-modality first
Lat–lon as the loss grid	Loss and writeback stay on <code>nCells</code> / deep levels

One-line lineage: *LandSim workflow + GraphCast residual mesh GNN, specialized to MPAS deep-ocean restart writeback.*

4. Architecture (as implemented)

4.1 System diagram

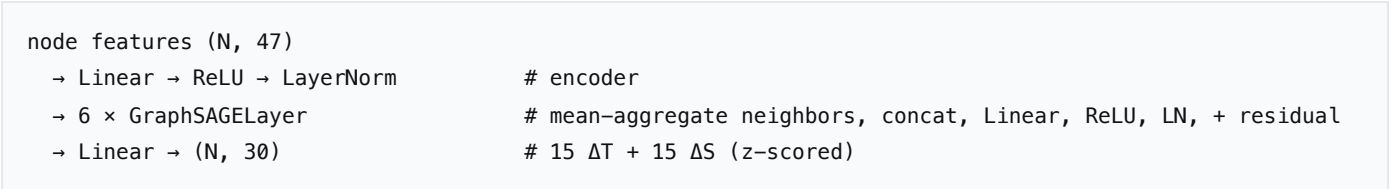


Package: oceanai/ (run_prototype.py stages the pipeline). Dependencies: numpy, torch, netCDF4 only — **no** PyG, **no** JAX GraphCast import.

4.2 Graph

Item	Value
Nodes	7153 cells
Edges	~41 018 directed edges from cells0nCell (max degree 6)
Node types (v1)	Homogeneous cells only (edge normalVelocity deferred)
Batching	One full mesh graph per pair (not mini-batches of cells)

4.3 Network (ResidualGNN)



Defaults used for published metrics: **hidden = 64, 6 layers**, Adam lr = 1e-3, ~15–25 epochs on CPU, checkpoint on best holdout deep-T RMSE.

4.4 Node features (47 channels)

Block	Width	Content
Early deep T	15	z-scored temperature at k=46...60
Early deep S	15	z-scored salinity
Static / season	7	sin/cos(lat), sin/cos(lon), log depth, month sin/cos
DATM month	10	u_10,v_10,t_10,slp_,q_10,dn10,prc,lwdn,swdn,swup for that month

Physics namelist (mpaso_variables) is stored as run metadata; **not** FiLM-conditioned in the shipped net.

4.5 Loss

Area $\times$ X-layerThickness weighted **Huber** on the concatenated normalized (ΔT , ΔS), zeroed where the deep mask is invalid (~63 228 valid deep points).

5. Data preparation

5.1 Source data (Kang / Frontier)

Stream	Role
Restarts years 051–055	Early state X
Restarts years 601–605	Late state Y
DATM CORE2 NYF (remapped to oQU240)	Monthly atmosphere on cells
mpaso_in / mpaso_variables	Static physics regime (one case)
History OHC AM	Validation only (not writeback)

Raw dump (~19 GB) stays on Lustre. Training off Frontier uses the **~113 MB AI-ready pack** (oceanai.data.pack_aiready).

5.2 Pair factory

Rule	Value
Alignment	Same calendar month, $\Delta = 550$ yr
Count	60 pairs (0051–01→0601–01, ..., 0055–12→0605–12)
Train	Years 51–54 → 601–604 (48)
Holdout	Year 55 → 605 (12)
Join	Identity on cell index (same mesh)

Each pairs/YYYY-MM.npz holds (7153, 15) arrays: t_x,s_x,h_x,t_y,s_y,h_y,valid.

5.3 Deep mask

```
refBottomDepth[k] > 2000 and k ≤ maxLevelCell[i] and bottomDepth[i] > 2000
```

→ **k = 46...60** ($\approx 2075\text{--}5500$ m), $N_{\text{DEEP}} = 15$.

5.4 Atmosphere encoding

Prefer **DATM state** (Tier-1 winds/T/SLP + precip + radiation), not coupler fluxes from ocean history (those depend on SST/ice). NYF **cycles**, so (F) is the same annual cycle every year — useful as context, not as a causal contrast in this dataset.

6. Predicted variables (v1 restart contract)

6.1 Written by the model

NetCDF name	Dimensions (written)	Levels	Units	How
temperature	nCells × deep	k = 46...60	°C	(T_X +)
salinity	nCells × deep	k = 46...60	1e-3	(S_X +)

Two prognostic fields, fifteen levels each → 30 output channels per cell. Only valid deep points are overwritten; inactive cells/levels keep the template.

6.2 Copied unchanged from the early restart (X)

Field	Why frozen in v1
temperature / salinity for k = 1...45	Upper ocean not in the deep-spinup target
layerThickness	Mass/volume from template; OHC uses X thickness
normalVelocity, normalBarotropicVelocity	Edge C-grid; deferred
Mesh connectivity, xtime	Keep early date on the AI restart

There is **no standalone ssh** in this restart schema; do not invent one.

6.3 Conditioning / metadata (not restart prognostics)

Input class	Examples	Role
DATM	u_10, v_10, t_10, slp_, prc, ...	Node features (monthly)
Geometry	areaCell, lat/lon, bottomDepth	Graph + static features
Namelist	GM/Redi κ, KPP, viscosity, ...	Run metadata (constant here)

6.4 Validation-only (history)

Field	Use
timeMonthly_avg_oceanHeatContent2000mToBot	Compare to restart-derived deep OHC
Maps / RMSE tables	Holdout interpretation

7. Pilot results (holdout January 0055-01 → 0605-01)

Metrics are **area × layer-thickness weighted** on the deep mask. Full snapshot:
prototype/snapshots/0055-01/compare.md.

7.1 Headline comparison

Metric	X (persistence)	ML	Y (truth)
Mean deep T (°C)	1.448	0.105	0.075
Deep T RMSE vs Y (°C)	1.433	0.392	—
Deep T bias (this – Y) (°C)	+1.373	+0.029	—
Mean deep S	34.736	34.726	34.714
Deep S RMSE vs Y	0.073	0.069	—
Deep S bias (this – Y)	+0.022	+0.012	—
OHC 2000 m–bot (J)	8.180×10^{26}	8.140×10^{26}	8.145×10^{26}
OHC rel. error vs Y	0.43%	0.065%	—
Shallow T/S, thickness, velocity vs X	—	0 (copy)	—

Reading: temperature skill is mostly a **large-scale cooling** toward Y. Salinity barely moves over 550 years — do **not** over-claim S skill. The writeback **contract** (unchanged non-deep fields) holds.

7.2 Mean deep temperature by level (°C)

k	z_ref (m)	X	ML	Y	ML – Y	X – Y
46	2075	2.229	1.017	0.983	+0.034	+1.246
47	2298	2.041	0.811	0.792	+0.019	+1.249
48	2530	1.865	0.619	0.605	+0.014	+1.260
49	2768	1.702	0.419	0.412	+0.007	+1.290
50	3011	1.559	0.245	0.218	+0.027	+1.341
51	3256	1.434	0.081	0.035	+0.046	+1.399
52	3503	1.318	–0.077	–0.131	+0.054	+1.449
53	3752	1.200	–0.219	–0.281	+0.062	+1.481

k	z_ref (m)	X	ML	Y	ML – Y	X – Y
54	4001	1.071	–0.371	–0.412	+0.041	+1.483
55	4251	0.930	–0.498	–0.526	+0.028	+1.456
56	4500	0.818	–0.602	–0.620	+0.017	+1.437
57	4750	0.754	–0.677	–0.694	+0.017	+1.448
58	5000	0.727	–0.726	–0.739	+0.012	+1.466
59	5250	0.740	–0.760	–0.758	–0.001	+1.498
60	5500	0.800	–0.758	–0.764	+0.006	+1.564

At every deep level, persistence (X–Y) stays ~1.2–1.6 °C too warm; ML–Y is typically a few hundredths of a degree.

7.3 Spatial maps (slides)

Field	Files
Column-mean deep T	prototype/snapshots/0055–01/map_T_{X,Y,ML,ML_minus_Y}.svg
Column-mean deep S	prototype/snapshots/0055–01/map_S_{X,Y,ML,ML_minus_Y}.svg (regenerate with python -m oceanai.qc.snapshot if missing)

8. Honest limitations (say these in the talk)

1. **One mesh, one NYF case, 48 train graphs** — easy to overfit; holdout is the next year of the same seasonal cycle.
2. **Salinity signal is weak** — RMSE improvement is small.
3. **No edge velocity, no upper ocean, no SSH** in v1.
4. **No N-day MPAS forward** in the pilot packet (Hyun after handoff).
5. **Do not** train GenCast/diffusion on these 60 pairs; **do not** lat–lon interpolate ocean state for a vision backbone.

9. How to use this with the slides

Slide deck	This document
docs/handoff_briefing.html	Live short talk (workflow, model card, tables, maps)
This file	Handout / speaker notes: lineage, IO contract, per-level table, caveats
tutorials/00–concepts.md	Vocabulary for implementers

Slide deck	This document
OceanAISpinup_Diffusion_Evaluation.md	Why not GenCast-first
OceanAISpinup_Development_Plan.md	Deeper GraphCast/LandSim code review

Suggested talk arc: problem → LandSim/GraphCast lineage → design principles → data pairs & deep mask → predicted vs copied variables → architecture box → comparison table → per-level T → maps → limitations / next owners.

10. References in-repo

- Code: `oceanai/models/residual_gnn.py`, `oceanai/io/write_restart.py`, `oceanai/run_prototype.py`
- Snapshot numbers: `prototype/snapshots/0055-01/compare.md`
- Variable selection: `data/docs/RESTART_DEEP_OCEAN_STATE.md`, `DATM_FORCING_VARIABLES.md`, `MPASO_PHYSICS_CONFIG_INPUTS.md`
- Run card: `prototype/README.md`